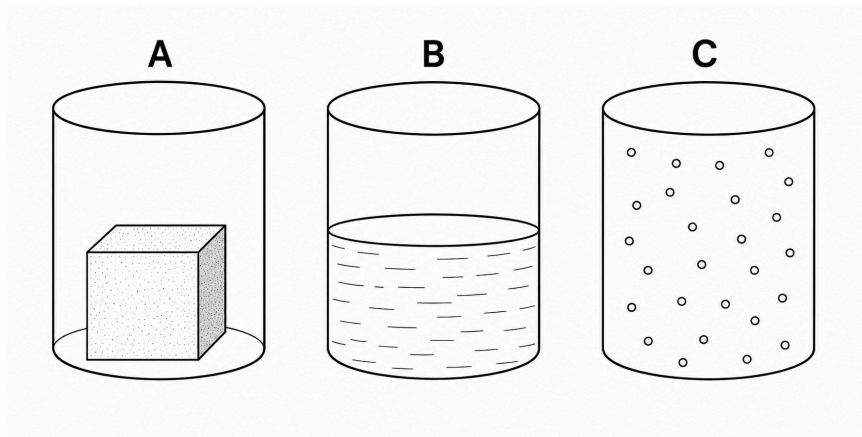


Criterion A -Knowing and understanding

Q1 A student has three substances:

- A has a fixed shape and fixed volume.
- B has a fixed volume but takes the shape of its container.
- C has no fixed shape or fixed volume.



Select the states of matter of A, B and C

Strand i | Achievement Level 1-2

States of matter

- **A – Solid**
- **B – Liquid**
- **C – Gas**

A solid has a fixed shape and fixed volume. A liquid has a fixed volume but takes the shape of its container. A gas has neither a fixed shape nor a fixed volume

Q2 Recall your scientific knowledge about particles in solids and gases.

a) Draw a particle diagram to show the arrangement and spacing of particles in a solid and in a gas.

b) State how the movement of particles differs in a solid and a gas.

Strand i | Achievement Level 3-4

a) Particle arrangement

Solid: Particles are closely packed in a fixed, regular arrangement, with very little space between them.

Gas: Particles are far apart, with large spaces between them and a random arrangement.

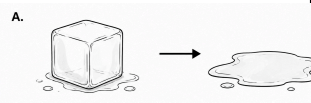
b) Particle movement

In a solid, particles vibrate about fixed positions. In a gas, particles move freely, rapidly and randomly in all directions.

Scientific terminology: closely packed, fixed positions, vibrate, far apart, random motion

Q3 State whether each image represents a physical or chemical change. Give a scientific reason for each answer.

Strand i | Achievement Level 5-6

Image	Type of Change	Scientific Reason
A. 	Physical change	The ice changes from a solid to a liquid , but no new substance is formed . It is still water.
B. 	Chemical change	Burning produces new substances , such as ash, smoke and gases . The original paper cannot easily be recovered.

Scientific terminology: physical change, chemical change, new substance, change of state

Q4 Dry ice changes directly from a solid into a gas. Using the particle model, **outline** what happens to the movement, arrangement and energy of the particles during this change. Name the change of state.

Strand i | Achievement Level 7-8

During sublimation, the particles absorb thermal energy and gain kinetic energy. They move faster and overcome the forces of attraction holding them in fixed positions. The particles change from being closely packed in a fixed arrangement to being far apart and randomly arranged. As a gas, they move rapidly and freely in all directions.

Scientific terminology: sublimation, thermal energy, kinetic energy, forces of attraction, particle arrangement

Q5 Two students want to keep their juice cold during a picnic.

Student A places the juice bottle in a container with ice.

Student B places the juice bottle in an empty container.

Select which method would keep the juice cold for longer and **suggest** why.

Strand ii | Achievement Level 1-2

Student A's method will keep the juice cold for longer.

The ice is at a lower temperature and absorbs thermal energy from the juice and surroundings as it melts. Therefore, the juice gains heat more slowly and remains cold for longer.

Scientific terminology: thermal energy, heat transfer, melting

Q6 A student removes a very cold bottle of water from the refrigerator. After a few minutes, small water droplets appear on the outside of the bottle.

Apply your knowledge of changes of state to:

- a) identify the change of state that produces the droplets;
- b) suggest why the droplets form on the cold surface.

Strand ii | Achievement Level 3–4

a) The

change of state is condensation.

b) Water vapour in the surrounding air comes into contact with the cold bottle and loses thermal energy. The particles lose kinetic energy, move more slowly and come closer together, changing from a gas into liquid water droplets.

Scientific terminology: condensation, water vapour, thermal energy, kinetic energy, gas → liquid

Q7 Two wet T-shirts are washed at the same time.

- T-shirt A is placed outside on a warm, windy day.
- T-shirt B is placed indoors in a cool, still room.

Apply particle theory to determine which T-shirt will dry faster. **Support your answer scientifically** using temperature and air movement.

Strand ii | Achievement Level 5–6

T-shirt A will dry faster.

On a warm day, the water particles gain more kinetic energy. More particles at the surface have enough energy to overcome the forces of attraction and escape into the air as water vapour. This increases the rate of evaporation.

The moving air also carries away water vapour from around the T-shirt, allowing more water to evaporate.

Therefore, higher temperature + greater air movement = faster evaporation.

Scientific terminology: evaporation, kinetic energy, surface particles, forces of attraction, water vapour, rate of evaporation

Q8 A shop needs to transport ice cream to a school picnic. The journey takes one hour, but there is no refrigerator in the vehicle.

a) Explain what happens to the movement and energy of the particles in the ice cream as it becomes warmer and begins to melt.

b) Suggest a way to pack the ice cream so that it remains solid for longer. Justify your solution using your knowledge of melting and particle theory.

Strand ii | Achievement Level 7-8

a) As the ice cream becomes warmer, its particles absorb thermal energy and gain kinetic energy. They vibrate more rapidly. When enough energy is absorbed, the particles overcome some of the forces of attraction holding them in fixed positions and begin to move past one another. The ice cream changes from solid to liquid. This process is called melting.

b) Pack the ice cream in an insulated container/cool box surrounded by ice or ice packs.

The insulation reduces heat transfer from the warmer surroundings to the ice cream. Therefore, the particles gain thermal energy more slowly and the ice cream takes longer to reach its melting point, so it remains solid for longer

Scientific terminology: thermal energy, kinetic energy, melting, melting point, heat transfer, insulation, forces of attraction

Q9 A student heats water and records the following observation:

Before Heating	After continued heating
Liquid water	Bubbles form throughout the water and water vapour is produced

The student says:

"The water is boiling and changing from a liquid into a gas."

Use the information in the table to make a **simple judgment** about whether the student's statement is correct. Give one piece of evidence from the table.

Strand iii | Achievement Level 1-2

The student's statement is correct.

The evidence shows that bubbles form throughout the water and water vapour is produced. This indicates boiling, in which liquid water changes into a gas

Scientific terminology: boiling, water vapour, liquid → gas, change of state

Q 10 A student observes two changes.

Change	Observation
Ice Melting	Liquid water forms and no new substance is produced
Wood Burning	Ash, smoke and gases are produced

A student says:

"Both are physical changes because matter is changing in both cases."

Apply the information in the table to make a judgment about whether the student's statement is correct. Give a scientific reason for your judgment

Strand iii | Achievement Level 3-4

The student's statement is incorrect.

Ice melting is a physical change because liquid water forms and no new substance is produced.

Only the state of matter changes.

Wood burning is a chemical change because new substances such as ash, smoke and gases are produced.

Therefore, although matter changes in both examples, they are not both physical changes.

Scientific terminology: physical change, chemical change, new substance, change of state

Q 11 Two identical gummy bears are placed in equal amounts of water.

Set Up	Condition
A	cold water
B	Warm Water

After the same amount of time, students observe that the gummy bear in warm water becomes softer and starts breaking down faster than the gummy bear in cold water.

a) Apply your knowledge of temperature and particle movement to explain why the gummy bear changes faster in warm water.

b) A student wants to slow down the change in a gummy bear placed in water. Suggest what the student could do and justify your suggestion using your knowledge of temperature and particle movement.

Strand iii | Achievement Level 5–6

a) The gummy bear changes faster in warm water because particles in warm water have greater kinetic energy and move faster. The faster-moving water particles interact with the gummy bear more frequently, so water moves into and dissolves/disperses parts of the gummy bear more quickly. Therefore, it becomes softer and breaks down faster.

b) To slow down the change, place the gummy bear in colder water.

At a lower temperature, the water particles have less kinetic energy and move more slowly.

Therefore, the interactions between the water and gummy bear occur more slowly, reducing the rate of change.

Scientific terminology: temperature, kinetic energy, particle movement, rate of change

Q12 A student places equal amounts of water in two dishes.

Dish	Temperature	Air movement	Water remaining after 2 hours
A	20°C	Still air	40 mL
B	35°C	Moving air	15 mL

The student concludes:

“The results prove that both higher temperature and moving air independently increase the rate of evaporation.”

Interpret the information and make a scientifically supported judgment about whether the evidence is sufficient to support the student's conclusion.

In your answer:

- use evidence from the table;
- use particle theory;
- make an overall scientifically supported judgment.

Strand iii | Achievement Level 7-8

The evidence shows that Dish B had a greater rate of evaporation because only 15 mL of water remained after 2 hours, compared with 40 mL remaining in Dish A. Dish B was at a higher temperature (35°C) and had moving air, while Dish A was at 20°C with still air.

According to particle theory, at a higher temperature water particles have greater kinetic energy, so more surface particles have enough energy to overcome the forces of attraction and escape into the air as water vapour. Moving air also removes water vapour from above the surface, which can increase the rate of evaporation.

However, the evidence is not sufficient to prove that higher temperature and moving air independently increase evaporation because two variables were changed at the same time: temperature and air movement. Therefore, we cannot determine from this experiment which factor caused the greater evaporation or how much each factor contributed.

To test the factors independently, the student should change only one variable at a time while keeping the other variables controlled.

Overall judgment: The results support the idea that the combination of higher temperature and moving air increased evaporation, but they do not independently prove the effect of each factor.

Key scientific terminology: evaporation • kinetic energy • particle theory • forces of attraction • water vapour • independent variable • controlled variable • sufficient evidence • scientifically supported judgment